

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/MANAGEMENT
/COMMERCIAL PRACTICE**

**INTRODUCTION TO MULTIMEDIA
MODEL QUESTION PAPER**

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	_____ refers to any type of application or presentation that involves more than one type of media, such as text, graphics, video, animation, and sound.	M1.01	R
2	You are preparing a multimedia presentation to be posted on your organization's website. You want to include images that you have gathered from the internet. What legal restriction must you consider before including these images? a) Copyright b) Relevance c) Resolution d) size	M1.02	R
3	Hypertext is a) used to link document b) a search engine c) includes sound and video d) requires plug-in to be viewed by a browser	M2.02	R
4	MIDI (Musical Instrument Digital Interface) is used by the multimedia computer to produce the music. Whether it is true or false.	M2.04	R
5	_____ audio/video refers to on-demand requests for compressed audio/video files. a) Streaming live b) Streaming stored c) Interactive d) none of the above	M2.03	R
6	_____ is used to compress images. a) MPEG92 b) JPEG c) either a or b d) none of the above	M3.04	R
7	When a program assigns 8 bits to a pixel, that pixel can display one of up to _____ colors.	M3.01	R
8	When a video clip merges and transforms into another image it is known as _____.	M4.03	R
9	Which one of these is a video file a) frogs.avi b) horse.jpg c) cat.mp3 d) All of the above	M4.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module outcome	Cognitive level
1	What is meant by the terms Multimedia and Hypermedia? Distinguish between these two concepts	M1.01	U
2	Interactivity is among the most important feature in any multimedia applications. What is interactivity and briefly explain what makes an application interactive?	M1.02	U
3	What key issues or problems does a multimedia system have to deal with when handling multimedia data?	M1.04	U
4	What is the mechanism of digitized sound?	M2.03	U
5	What is the advantage of MIDI over digitized sound?	M2.04	U
6	How does the computer reconstruct sound wave from a sample data?	M2.04	U
7	Explain the TIFF image compression standard.	M3.04	U
8	Summarize various advantages of image compression.	M3.03	U
9	Show why delta modulation is classified as lossless compression.	M3.02	U
10	What are the 3 characteristics of colour? What kind of colour does a computer display.	M4.01	U

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

		Module outcome	Cognitive level
1.	What issues of functionality need to be provided in order to effectively use a wide variety of media in Multimedia applications?	M1.04	U
	OR		
2.	Explain the use of the following hardware components in multimedia. (i) Scanner (ii) Soundcard	M1.04	U
3.	What causes sound? How is sound recorded digitally?	M2.03	U
	OR		
4.	Why is data compression necessary for Multimedia activities? What is the distinction between lossless and lossy compression?	M2.05	U
5.	With the help of example explain the method of dictionary based coding.	M2.05	U
	OR		
6.	Demonstrate the concept of sliding window compression method.	M2.05	U

7.	Explain the architecture of JPEG in detail.	M3.04	U
	OR		
8	What are still images? Describe various ways of making still images.	M3.01	U
9.	Explain various computer colour models in detail.	M3.01	U
	OR		
10.	Explain how Huffman coding can be employed in image compression.	M3.03	U
11.	Explain MPEG motion video compression standard in detail.	M4.02	U
	OR		
12.	Explain multimedia distribution network ATM in detail.	M4.04	U